

Capacitor Charging and a Power On Reset for an Active Low Microcontroller

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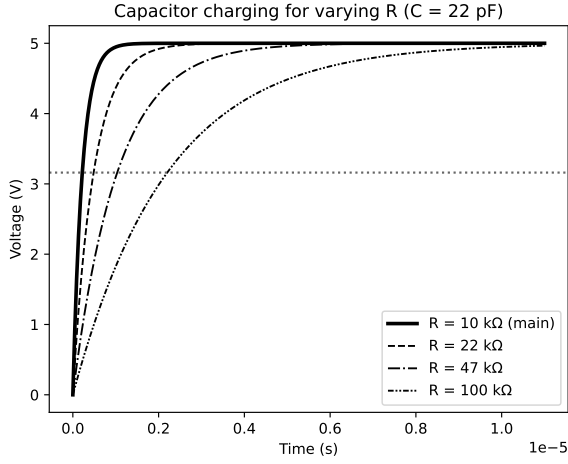


Fig. 1. Capacitor charging for different resistance values.

Abstract—This document introduces the charging behaviour of a resistor-capacitor (RC) network, examines the capacitor voltage at one time constant, and applies the same charging behaviour to a power-on-reset circuit for an active-low microcontroller.

I. INTRODUCTION

When a capacitor C is charged through a series resistor R from a constant supply V_0 , the capacitor voltage rises according to

$$V(t) = V_0 \left(1 - e^{-t/RC}\right)$$

where V_0 is the supply voltage, R is the resistance, and C is the capacitance. The product RC is called the time constant,

$$\tau = RC$$

After one time constant, the capacitor reaches approximately 63.2% of the supply voltage.

II. VOLTAGE AT THE TIME CONSTANT

Starting from the charging equation,

$$V(t) = V_0 \left(1 - e^{-t/RC}\right)$$

and substituting $t = \tau$,

$$V(\tau) = V_0 \left(1 - e^{-\tau/RC}\right)$$

Since $\tau = RC$,

$$V(\tau) = V_0 \left(1 - e^{-RC/RC}\right)$$

$$V(\tau) = V_0 \left(1 - e^{-1}\right)$$

$$V(\tau) \approx 0.632V_0$$

For $V_0 = 5.0$ V,

$$V(\tau) \approx 0.632 \times 5.0$$

$$V(\tau) \approx 3.16$$
 V

III. POWER ON RESET WITH AN ACTIVE LOW RESET

A power-on-reset circuit can be used to hold a microcontroller in reset while the supply voltage is first turning on. In this circuit, the capacitor charges through the resistor. The active-low reset pin remains low while the capacitor voltage is below the input-high threshold V_{IH} . Once the capacitor voltage crosses V_{IH} , the reset is released and the microcontroller can begin operating.

The reset pin releases when the node voltage reaches V_{IH} . Starting from

$$V_{IH} = V_0 \left(1 - e^{-t_{\text{release}}/RC}\right)$$

divide by V_0 :

$$\frac{V_{IH}}{V_0} = 1 - e^{-t_{\text{release}}/RC}$$

Rearranging,

$$e^{-t_{\text{release}}/RC} = 1 - \frac{V_{IH}}{V_0}$$

Taking the natural logarithm,

$$-\frac{t_{\text{release}}}{RC} = \ln \left(1 - \frac{V_{IH}}{V_0}\right)$$

Therefore,

$$t_{\text{release}} = -RC \ln \left(1 - \frac{V_{IH}}{V_0}\right)$$

Solving for R gives

$$R = \frac{-t_{\text{release}}}{C \ln \left(1 - \frac{V_{IH}}{V_0}\right)}$$

For the worked example, let

$$V_{IH} = 0.7V_0$$

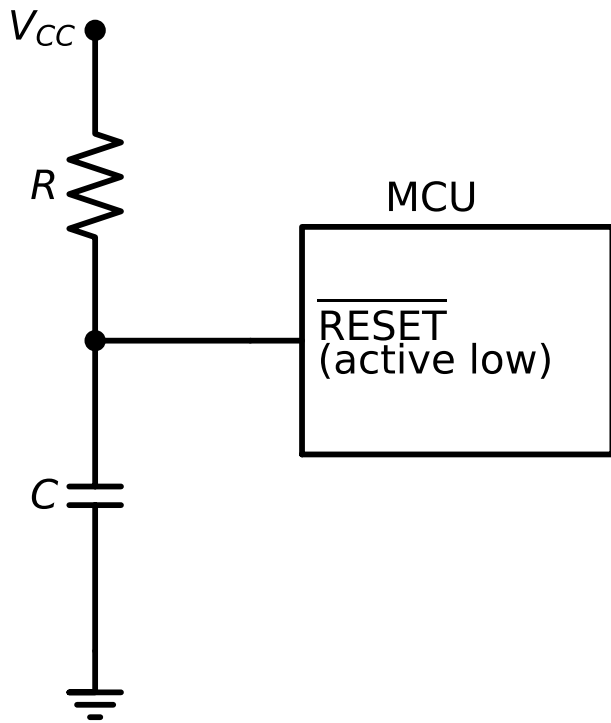


Fig. 2. RC power-on-reset network driving an active-low reset pin.

with $V_0 = 5.0 \text{ V}$, $t_{\text{release}} = 10 \text{ ms}$, and $C = 100 \text{ nF}$.

$$\ln(1 - 0.7) = \ln(0.3) \approx -1.204$$

Therefore,

$$R = \frac{-(10 \times 10^{-3})}{(100 \times 10^{-9})(-1.204)}$$

$$R \approx 83 \text{ k}\Omega$$

An $83 \text{ k}\Omega$ resistor with a 100 nF capacitor gives a reset hold-off of approximately 10 ms .